

Problem 5

$$C_{\text{ppm}} = 0.0825 \frac{T}{PM} (\text{mg/m}^3)$$

$$(\text{mg/m}^3) = \frac{C_{\text{ppm}} PM}{0.0825 T}$$

$$(\text{mg/m}^3) = \frac{1 \cdot 1 \cdot 70.9}{0.08205 \cdot 298.15}$$

$$(\text{mg/m}^3) = 2.89$$

Assume F-stability, u=2ms/, rural

$$\sigma_y = 0.04x (1 + 0.0001x)^{-1/2}$$

$$\sigma_y = 0.04 \cdot 300 (1 + 0.0001 \cdot 300)^{-1/2}$$

$$\sigma_y = 11.8$$

$$\sigma_z = 0.016x (1 + 0.0003x)^{-1}$$

$$\sigma_z = 0.016 \cdot 300 (1 + 0.0003 \cdot 300)^{-1}$$

$$\sigma_z = 4.4$$

$$C = \frac{Q_m}{\pi \sigma_y \sigma_z u}$$

$$Q_m = C \pi \sigma_y \sigma_z u$$

$$Q_m = 2.89 \cdot \pi \cdot 11.8 \cdot 4.4 \cdot 2$$

$$Q_m = 948 \text{ mg/s}$$

$$Q_m = C_0 A P \sqrt{\frac{\gamma g_c M}{R_g T} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}$$

$$\frac{948}{1000000} = 1 \cdot A \cdot 51 \cdot 10^5 \sqrt{\frac{1.33 \cdot 1 \cdot 70.9}{8.314 \cdot 1000 \cdot 298} \left(\frac{2}{1.33 + 1} \right)^{\frac{1.33+1}{1.33-1}}}$$

$$A = 5.166 \cdot 10^{-8} \text{ m}^2$$

$$A = \frac{\pi}{4} D^2$$

$$D = \sqrt{\frac{4A}{\pi}}$$

$$D = \sqrt{\frac{4 \cdot 5.166 \cdot 10^{-8}}{\pi}}$$

$$D = 2.56 \cdot 10^{-4} \text{ m}$$

$$\boxed{D = 0.256 \text{ mm}}$$